

Bayesian Estimation of the Probability of Informed Trading (PIN) on Polymarket

Model, Results, and Conclusions

2,263 resolved markets, ~29M trades • Sept 2024 – Feb 2025

Politics (Trump vs Harris), Fed rate cuts (50bp Sept → pause Jan), Crypto (BTC past \$100K), Sports (NFL playoffs, Super Bowl LIX)

Q1. Does Bayesian estimation improve over MLE for PIN?

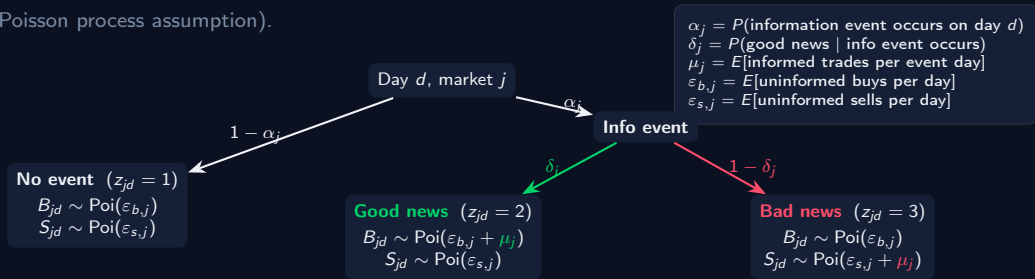
Q2. Do the four specifications (M1–M4) give different answers?

Q3. Does estimated PIN tell us anything useful about price accuracy?

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The Data Generating Process (Easley et al. 1996)

Five parameters per market j . Each day, nature picks one of three **latent day types**: (1) no information event; (2) **good news**; (3) **bad news**. Then trades arrive independently at constant rates within each day according to day type (Poisson process assumption).



Data: 2,263 resolved markets, $\sim 29\text{M}$ trades (Sept 2024 – Feb 2025). B_{jd} = (buy YES + sell NO), S_{jd} = (sell YES + buy NO) per day.
 Filters: ≥ 20 days, ≥ 500 trades. Final 24h excluded. Median duration: 57 days. 24.9% of markets have < 30 days.

$$\text{PIN}_j = \frac{\alpha_j \mu_j}{\alpha_j \mu_j + \varepsilon_{b,j} + \varepsilon_{s,j}} \quad (\text{derived from posterior draws, not a parameter — no prior, no likelihood of its own})$$

Notation:

j = market d = trading day $z_{jd} \in \{1, 2, 3\}$ = day type B_{jd}, S_{jd} = counts, B_{jd}^i = imputed (Slide 6)

Specification-specific indices: M2: k = category (all params) M3: r = regime (α only) M4: t = time (α only)

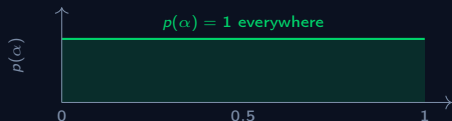
The DGP: Priors (also M1 default priors)

These priors are deliberately vague to express near-ignorance about parameter values before seeing data.

Param	Support	Prior*	PDF	Why
α_j	$[0, 1]$	Beta(1, 1)	$p(\alpha) = 1$ (flat)	Equal weight on all values in $[0, 1]$
δ_j	$[0, 1]$	Beta(1, 1)	$p(\delta) = 1$ (flat)	Equal weight on all values in $[0, 1]$
μ_j	$[0, \infty)$	Gamma(1, 0.01)	$p(\mu) = 0.01e^{-0.01\mu}$	Mean = 100, SD = 100: allows 1–10K
$\varepsilon_{b,j}$	$[0, \infty)$	Gamma(1, 0.01)	$p(\varepsilon_b) = 0.01e^{-0.01\varepsilon_b}$	Same: wide enough for any market
$\varepsilon_{s,j}$	$[0, \infty)$	Gamma(1, 0.01)	same	Same as ε_b

*Shared across all four specifications. M2 replaces these with hierarchical priors learned from category data (details on M2 slide).

Beta(1,1) — prior for α_j, δ_j



Gamma(1, 0.01) — prior for $\mu_j, \varepsilon_{b,j}, \varepsilon_{s,j}$



Flat: $\alpha = 0.01$ and $\alpha = 0.99$ are equally likely *a priori*.
Barely constrains: $\mu = 5$ and $\mu = 500$ both plausible *a priori*.
After 20–30 days of (B, S) count data, the likelihood overwhelms these priors — posteriors are driven almost entirely by observed data.

The DGP: Four Structural Assumptions About α

All four specifications share the same likelihood. They differ in how α behaves (and in M2's case, all parameters).

Spec.	Assumption about α_j	Prior on α_j	Other parameters
M1	α_j is constant across all days. Each market independent (No borrowing across markets).	$\alpha_j \sim \text{Beta}(1, 1)$	$\delta_j, \mu_j, \varepsilon_{b,j}, \varepsilon_{s,j}$: flat/vague priors from Slide 3.
M2	α_j is constant, but similar within category (e.g. politics, sports). Thin markets borrow from data-rich ones.	$\log(\alpha_j) \sim N(m_k, s_k^2)$	All five get hierarchical priors: $\log(\mu_j) \sim N(m_{\mu,k}, s_{\mu,k}^2)$, $\log(\varepsilon_{b,j}) \sim N(m_{\varepsilon_b,k}, s_{\varepsilon_b,k}^2)$, etc. Category means m_k and variances s_k^2 learned from data.
M3	α_j switches between α_{quiet} and α_{active} (two info event rates). Quiet: few events. Active: frequent.	$\alpha_{j,r} \sim \text{Beta}(1, 1)$ per regime	$\delta_j, \mu_j, \varepsilon_{b,j}, \varepsilon_{s,j}$: same as M1. New: $p_{qq} \sim \text{Beta}(9, 1)$, $p_{aa} \sim \text{Beta}(9, 1)$. Beta(9,1) encodes sticky regimes (persist over days).
M4	α_j drifts daily via Gaussian noise on logit scale. Info frequency changes continuously.	$\text{logit}(\alpha_{j,t}) = \text{logit}(\alpha_{j,t-1}) + \eta_t$, $\eta_t \sim N(0, \sigma_\eta^2)$	$\delta_j, \mu_j, \varepsilon_{b,j}, \varepsilon_{s,j}$: same as M1. New: $\sigma_\eta^2 \sim \text{IG}(2, 1)$. Small $\sigma_\eta^2 \rightarrow$ collapses to M1. Large $\sigma_\eta^2 \rightarrow$ mimics M3.

How does the information environment behave in prediction markets? We specify four competing hypotheses, each implying a different prior structure. We estimate each separately via MCMC, then compare fit using WAIC and Bayes factors.

The DGP: Joint Likelihood

Daily counts (B_{jd}, S_{jd}) are Poisson-distributed (standard count model). The likelihood sums over the 3 unobserved day types.

Poisson pmf: $f(k | \lambda) = \frac{e^{-\lambda} \lambda^k}{k!}$ — probability of k trades when expected rate is λ . Each f below is poisson pmf with B_{jd} or S_{jd} plugged in as k and their respective rates plugged in as λ .

Joint likelihood for market j , day d (sum because day type z_{jd} is unobserved):

$$L_{jd} = \underbrace{(1-\alpha_j) f(B_{jd}|\varepsilon_{b,j}) f(S_{jd}|\varepsilon_{s,j})}_{\text{no event: noise only}} + \underbrace{\alpha_j \delta_j f(B_{jd}|\varepsilon_{b,j} + \mu_j) f(S_{jd}|\varepsilon_{s,j})}_{\text{good news: } \mu_j \text{ extra buys}} + \underbrace{\alpha_j (1-\delta_j) f(B_{jd}|\varepsilon_{b,j}) f(S_{jd}|\varepsilon_{s,j} + \mu_j)}_{\text{bad news: } \mu_j \text{ extra sells}}$$

B_{jd} and S_{jd} are the **observed data**. On a good-news day, **buys** are evaluated at rate $\varepsilon_{b,j} + \mu_j$ (noise + informed) instead of $\varepsilon_{b,j}$ (noise only). Same logic for a bad-news day.

Full market likelihood: $L_j = \prod_{d=1}^{T_j} L_{jd}$. **This likelihood is the same for all four specifications (M1–M4).**

Bayesian Inference: Full Conditionals via Data Augmentation

Problem: Hard to isolate α_j to get $p(\alpha_j \mid \delta_j, \mu_j, \varepsilon_{b,j}, \varepsilon_{s,j}, \text{data})$. Solution: data augmentation (Griffin et al. 2021)

Data Augmentation: At each MCMC iteration, **impute** $z_{jd} \in \{1, 2, 3\}$ for every day:

$$P(z_{jd}=2 \mid B_{jd}, S_{jd}, \theta_j^{(s)}) = \frac{\alpha_j^{(s)} \delta_j^{(s)} \cdot f(B_{jd} \mid \varepsilon_{b,j}^{(s)} + \mu_j^{(s)}) \cdot f(S_{jd} \mid \varepsilon_{s,j}^{(s)})}{L_{jd}^{(s)}}$$

Using current parameters, compute how probable each type is for each day's (B_{jd}, S_{jd}) . Sample z_{jd} from these probabilities — each day gets a type for this iteration during MCMC. A single iteration's label may be wrong. But over thousands of iterations, the sampler explores all plausible configurations — uncertainty in day types propagates into parameter uncertainty.

Result for α_j : Count $n_{\text{event}} = \#\{d : z_{jd} \in \{2, 3\}\}$, $n_{\text{no}} = \#\{d : z_{jd} = 1\}$. Full conditional becomes:

$$p(\alpha_j \mid z, \text{data}) \propto \underbrace{\alpha_j^{n_{\text{event}}} (1 - \alpha_j)^{n_{\text{no}}}}_{\text{Beta kernel: count event vs non-event days}}$$

Trade split: $B_{jd}^i \sim \text{Binomial}(B_{jd}, \mu_j / (\mu_j + \varepsilon_{b,j}))$ separates informed from uninformed buys. Now μ_j counts informed trades (Gamma kernel), $\varepsilon_{b,j}$ counts uninformed (Gamma kernel). All five parameters decouple.

Bayesian Inference: Prior $\times$ Full Conditional $\propto$ Posterior

Every parameter across all four specifications. Same likelihood and augmentation throughout — only priors differ.

PIN model parameters (enter the Poisson likelihood):

	Param	Prior	Full cond. kernel		Posterior
M1	α_j	Beta(1, 1)	$\alpha^{n_{\text{ev}}} (1-\alpha)^{n_{\text{no}}}$	Beta	$\rightarrow$ Beta(1+ n_{ev} , 1+ n_{no}) [Gibbs]
	δ_j	Beta(1, 1)	$\delta^{n_{\text{good}}} (1-\delta)^{n_{\text{bad}}}$	Beta	$\rightarrow$ Beta(1+ n_{good} , 1+ n_{bad}) [Gibbs]
	μ_j	Ga(1, .01)	$\mu^{\sum \text{inf}} e^{-n_{\text{ev}} \mu}$	Gamma	$\rightarrow$ Ga(1+ $\sum \text{inf}$, .01+ n_{ev}) [Gibbs]
	$\varepsilon_{b,j}, \varepsilon_{s,j}$	Ga(1, .01)	$\varepsilon^{\sum \text{uninf}} e^{-T \varepsilon}$	Gamma	$\rightarrow$ Ga(1+ $\sum \text{uninf}$, .01+ T) [Gibbs]
M2	θ_j (all 5)	LogN($m_{\theta,k}, s_{\theta,k}^2$)	same kernels as M1	Beta/Gamma	$\rightarrow$ LogN $\times$ Beta/Gamma = ?? [MH]
M3	$\alpha_{j,r}$	Beta(1, 1)	$\alpha_r^{n_{\text{ev}}^{(r)}} (1-\alpha_r)^{n_{\text{no}}^{(r)}}$	Beta	$\rightarrow$ Beta(1+ $n_{\text{ev}}^{(r)}$, 1+ $n_{\text{no}}^{(r)}$) [Gibbs]
M4	$\alpha_j(t)$	$\ell_t \sim N\left(\frac{\ell_t - 1 + \ell_{t+1}}{2}, \frac{\sigma_\eta^2}{2}\right)$	$\alpha_t^{\mathbb{1}^{\text{[ev]}}} (1-\alpha_t)^{\mathbb{1}^{\text{[not]}}}$	Beta	$\rightarrow$ LogitN $\times$ Beta = ?? [MH]

M4: $\ell_t = \text{logit}(\alpha_t)$. Prior is Normal on ℓ_t conditional on neighbors. Kernel is Beta from one day only ($n_{\text{ev}}, n_{\text{no}} \in \{0, 1\}$).

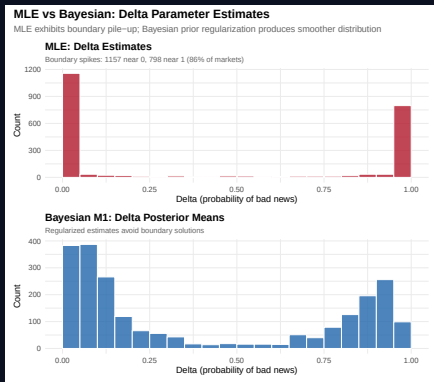
Specification-specific parameters (don't enter Poisson likelihood — updated given latent states/params):

	Param	Prior	Full cond. kernel		Posterior
M2	$m_{\theta,k}$ (categ. means)	$N(0, 10)$	$\prod_{j \in k} N(\log \theta_j \mid m_{\theta,k}, s_{\theta,k}^2)$	Normal	$\rightarrow N(\bar{\ell}_k, s_{\theta,k}^2/n_k)$ [Gibbs]
	$s_{\theta,k}^2$ (categ. vars.)	Inv-Ga(10, 1) tight Inv-Ga(2, 1) moderate Inv-Ga(0.5, 0.5) diffuse	$\prod_{j \in k} N(\log \theta_j \mid m_{\theta,k}, s_{\theta,k}^2)$	Inv-Gamma	$\rightarrow$ Inv-Gamma($\cdot, \cdot$) [Gibbs]
M3	p_{qq}, p_{aa} (stay probs)	Beta(9, 1)	$p^{n_{\text{stay}}} (1-p)^{n_{\text{switch}}}$	Beta	$\rightarrow$ Beta(9+ n_{stay} , 1+ n_{switch}) [Gibbs]
M4	σ_η^2 (drift var.)	Inv-Gamma(2, 1)	$(\sigma_\eta^2)^{-T/2} \exp\left(-\frac{\sum_t \eta_t^2}{2\sigma_\eta^2}\right)$	Inv-Gamma	$\rightarrow$ Inv-Gamma $\left(2+\frac{T}{2}, 1+\frac{\sum_t \eta_t^2}{2}\right)$ [MH]

Latent sequences sampled each iteration: M3 regimes $\{s_d\}$ (FFBS), M4 $\alpha(t)$ path (single-site MH). Not parameters.

Q1: Does Bayesian Estimation Improve Over MLE?

M1 (Static) vs MLE



Top: MLE δ piles up at exactly 0 and 1. In 86% of markets, MLE concludes information events were either *all* good or *all* bad.

Bottom: Bayesian posterior means spread across the full range with no boundary spikes. Estimates that most markets received more bad news but distribution is smoother.

	MLE	Bayes (M1)
Median PIN	0.274	0.292
δ boundary	84.4%	0%

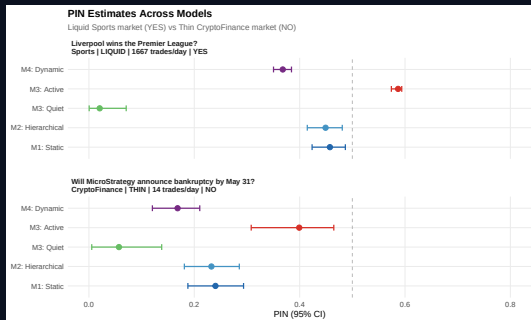
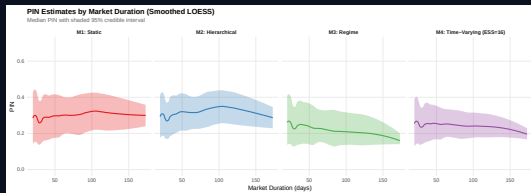
Why: $\hat{\delta}_{MLE} = n_{\text{good}} / n_{\text{event}}$. E.g., five event days all bad $\rightarrow 0/5 = 0$. MLE then concludes good-news events are *impossible* in this market. This locks the entire parameter vector into a constrained optimum, distorting PIN. Thin markets ($<30d$) face this most, because they don't have enough data: 89.2% boundary rate.

Bayes: $\text{Beta}(1+0, 1+5) \rightarrow \text{mean } 0.14, \text{ CI } [0.01, 0.47]$.
“Good news events are probably rare, but not impossible.”

Other parameters don't face the same problem (α boundary rate: 0.1%). The issue is specific to δ .

Q2: Do the four specifications (M1–M4) give different answers?

Comparison of PIN estimates across 4 models



Model	Median PIN	95% CI width
M1 (static)	0.292	0.202
M2 (hierarchical)	0.301	0.183
M3 (regime)	0.222	0.177
M4 (dynamic)*	0.240	0.162 [†]

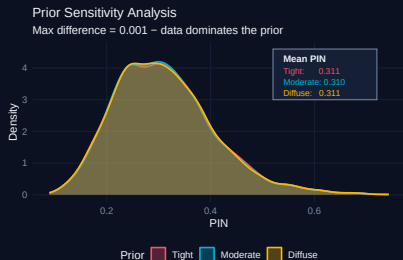
*ESS = 16 so CI is artificially narrow as chain underexplores.

CI shrinks over time. M1/M2 flat at ~ 0.30 - they estimate one α per market, so PIN doesn't vary with time. M3/M4 lower (~ 0.22) and trend down: since α changes over time, they find that longer markets spend more time in quiet periods as time passes.

For "Liverpool wins PL" (liquid), M1 gives one moderate estimate. M3 splits it into two tight clusters. Most days are quiet with very low PIN, active days very high PIN. For "MicroStrategy bankruptcy" (thin), all models give wide CIs. M2 and M1 are very similar for both markets.

Q2: Do the four specifications (M1–M4) give different answers?

M2: Does the hierarchy help, and does the choice of hyperprior matter?

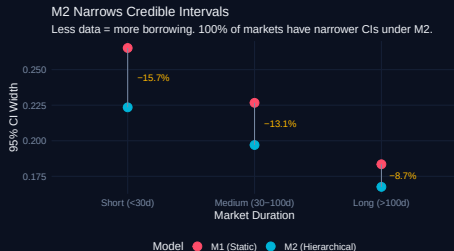


Hyperprior Sensitivity: Three hyperpriors on $s_{\theta,k}^2$ (tight, moderate, diffuse) produce nearly identical PIN distributions. With 2,263 markets, the data overwhelms the prior.

CI shrinkage: The hierarchy narrows CIs for every market, most where data is scarcest:

- Thin markets: 12.2% narrower CIs. Liquid: 8.3%
- <30d: 15.2% narrower. >100d: 5.2%
- 100% of markets have narrower CIs under M2

But: only 47% of markets are pulled *toward* the category mean (so the hierarchy is not doing classic shrinkage). This could be because M2 learns that within-category spread is large and accommodates it rather than forcing markets together.

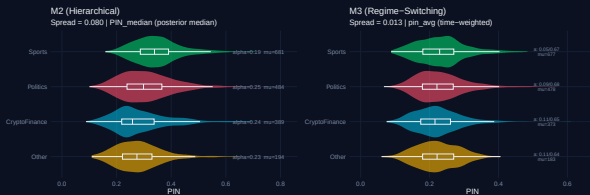


Q2: Do the four specifications (M1–M4) give different answers?

M2 finds category differences in PIN. M3 says those differences disappear once you account for regimes.

Category PIN Distributions: M2 vs M3

Politics: most info events ($\alpha=0.25$) | Sports: largest bursts ($\mu=681$) | Other: smallest ($\mu=194$) | M3 alpha format: quadratic



Left: M2 says Sports has higher PIN than other cats. Right: under M3, the differences nearly vanish, all PIN are ~ 0.22 .

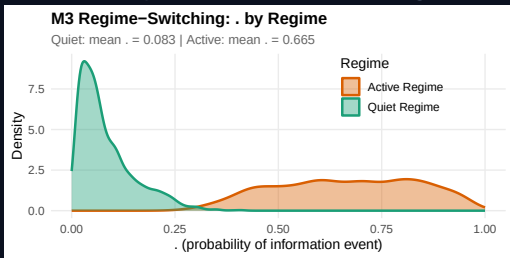
M2: Sports has the least info events ($\bar{\alpha} = 0.187$) but largest bursts of informed trading when they happen ($\bar{\mu} = 674$) Politics is opposite: most frequent info events ($\bar{\alpha} = 0.243$) but more moderate intensity ($\bar{\mu} = 488$). PIN combines both ($\alpha\mu/(\alpha\mu + \varepsilon_b + \varepsilon_s)$), so Sports' rare-but-huge bursts produce a higher static PIN than Politics' frequent-but-moderate ones.

M3: Once you separate quiet and active days, the difference vanishes. Every category has similar quiet-regime PIN (~ 0.11) and similar active-regime PIN (~ 0.55). Sports just switches into active mode more sharply (4.1 switches, $\Delta\text{PIN} = 0.52$) while Politics drifts in and out more gradually.

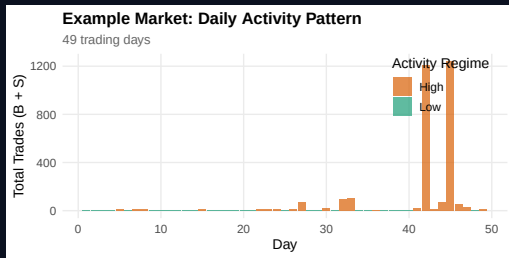
Categories differ in *when* and *how sharply* information arrives, not *how much*.

Q2: Do the four specifications (M1–M4) give different answers?

Did M3 find quiet vs active information regimes, and how does PIN differ?



Two clearly separated distributions: quiet α near 0.08, active around 0.67. There are **8× more** info events during active regimes, and this trend appears in **99.4%** of markets.



Example market (49 days). Orange = M3's active regime.

Trading volume visibly spikes during active periods, suggesting real informed trading activity

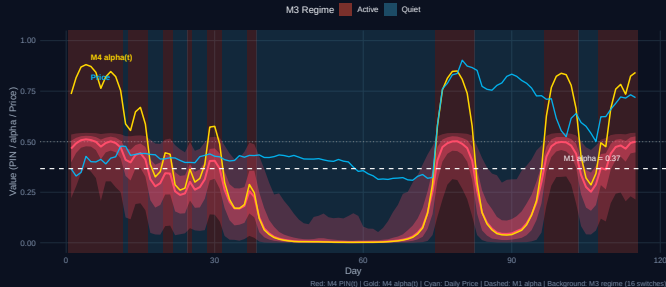
Regime dynamics: Markets spend **71.8%** of their life in quiet (~ 19 day spells) with short active bursts (~ 11 days). M1's single $\alpha = 0.22$ averages over both, so it would have overstated quiet alpha and understated active alpha. **M3 overall PIN is lower than M1 (0.222 vs 0.292)** — not less informed trading, but M3 correctly weights the fact that markets are quiet most of the time.

Q2: Do the four specifications (M1–M4) give different answers?

M4: Does Information Intensity Drift? Example: “Trump declassifies JFK files in first 100 days?”

M4 Dynamic PIN Path (115 days, ESS=23) → YES

Trump declassifies JFK files in first 100 days?



ESS = 23

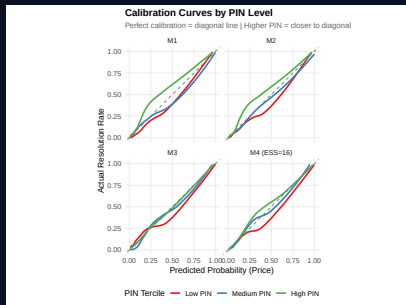
M4 samples one α_t per day via single-site MH — 115 correlated parameters. Chain moves slowly through this high-dimensional space, so draws are highly autocorrelated.

What M4 finds: α oscillates between 0.15 and 0.90 — information arrives in waves. Around days 55–80, a sustained burst of informed trading (gold line peaks) pushes the cyan price line sharply upward toward the correct YES resolution. M1’s flat dashed line at $\alpha = 0.37$ misses all of this.

M4 vs M3: Background shading shows M3’s 16 discrete regime switches. M4’s gold line shows the same bursts but with gradual buildup and decay; M3 treats transitions as instantaneous. Both models agree on *when* information arrives, validating the assumption that information flow is not constant.

Q3: Does estimated PIN tell us anything useful about price accuracy?

(b) When a market trades at 60¢, does the event resolve YES 60% of the time? How does PIN affect this?



Markets with high PIN track the 45° diagonal closely, reflects true probability. Markets with low PIN are less reliable forecasts. M3 (bottom left) shows tightest diagonal fit overall across all PIN.

Brier Score: $BS = \frac{1}{N} \sum (p_j - \omega_j)^2$

p_j = market price, $\omega_j \in \{0, 1\}$ = actual outcome.

Measures how far prices are from truth. Lower = more accurate. Perfect forecasts score 0, coin flips score 0.25.

Brier Score by PIN tercile (lower = better):

Model	Low	High	p
M1	0.075	0.060	$< 10^{-40}$
M2	0.082	0.057	$< 10^{-101}$
M3	0.076	0.049	$< 10^{-136}$
M4*	0.075	0.050	$< 10^{-120}$

All four models agree: markets with more informed trading produce better-calibrated prices. High PIN: 87–90% accurate vs 76–82% for low PIN.

Why? Uninformed traders on Polymarket bet YES 65% more more than NO ($\varepsilon_b/\varepsilon_s = 1.65$). But 83% of events resolve NO. Informed traders thus profit by selling against their systematic optimism, which actually helps push prices closer to truth.

M3 produces best calibration: Separating quiet from active periods might have produced PIN estimates that divided into cleaner meaningful groups.

Q3: Does estimated PIN tell us anything useful about price accuracy?

(c) We showed high PIN calibrates better prices. Now: which model's PIN is most useful for predicting accuracy?

WAIC: Which model best fits the trading data?

WAIC measures how well each model predicts the daily buy/sell counts it was trained on. Lower = better. Penalizes unnecessary complexity.

Model	WAIC ($\times 10^6$)
M1 (static)	12.20
M3 (regime)	12.95
M4 [†]	13.30
M2 [‡]	28.18

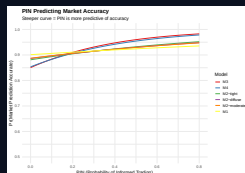
[†]ESS = 16. [‡]M2 hierarchical p_{WAIC} unstable. Not comparable.

M1 wins. M3's regimes improve fit, but not enough to justify the extra parameters. WAIC's complexity penalty favors the simpler model.

Key tension: M1 fits *trading data* best. M3 produces the most *useful* PIN. M3 sacrifices fit to separate quiet from active, thus making PIN informative.

Logistic: Which model's PIN best predicts whether a market's price is accurate?

"Accurate" = median price correctly predicts resolution. We regress accuracy on PIN: positive β_1 means higher PIN predicts more accurate markets.



Model	β_1	p
M3	2.88	0.013
M4 [†]	2.56	0.017
M2	1.23	0.130
M1	0.58	0.527

Steeper curve = PIN more predictive. M3 (red) rises fastest.

M3 wins. Only dynamic models' PIN significantly predicts accuracy ($p < 0.05$). Static PIN doesn't discriminate.

Three Questions, Three Answers

Q1. Does Bayesian improve over MLE? **Yes.**

MLE places δ at exact 0 or 1 in **84.4%** of markets (0% for Bayesian). The two converge for well-traded markets ($r = 0.80$), but Bayesian wins in thin markets where MLE breaks down.

Q2. Do the models differ? **Yes — they agree on how much, disagree on when.**

All models: **~22–30%** of trades are informed. M3 reveals **99.4%** of markets alternate between quiet (PIN = 0.11) and active (PIN = 0.55). Apparent category differences in PIN arise due to the different ways models treat burst intensity.

Q3. Does PIN predict accuracy? **Yes — high PIN means trustworthy prices.**

High-PIN markets: **87–90%** accurate vs **76–82%** low PIN. Informed traders exploit uninformed optimism *and* push prices toward truth. Practically: If we run M3 on live data, high PIN would signal that the market price reliably reflects the true probability of the event resolving YES.

The Bigger Picture: PIN as Cost vs. Quality

Equity markets (Easley & O'Hara 2004)

Informed exploit uninformed → uninformed leave →
liquidity dries up → market gets worse.

PIN = cost.

Prediction markets (this project)

Informed exploit uninformed → prices move toward
truth → everyone who *reads* the price benefits.

PIN = quality.

Same mechanism — informed profiting from uninformed — opposite implications depending on what the market is *for*.

The tension: Should platforms **protect uninformed traders** (less exploitation, but prices become noisier) or **encourage informed trading** (more exploitation, but better forecasts)?

Our results say high-PIN markets give the most accurate prices. The social value of prediction markets — reliable probability forecasts that journalists cite, hedge funds trade on, and policymakers consult — is funded by the people who lose money to traders with inside knowledge.